

POWERBOX · REQUIREMENTS

PowerBox requirements

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DOCUMENT CONTROL

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Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/powerbox		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	R2 restated as 1.8 evenings (Amish, 2026-10-02, PBX-DEC-001 item 5); met on paper
0.5	2026-10-01	Amish Chadha	R10 and R11 status for the constructable design (PBX-DDR-003, PBX-CAL-001 v0.2); budget treated as a value-engineering target
0.4	2026-09-25	Amish Chadha	TRL 3. Record Amish's decisions (PBX-DDR-001), including the 230 V AC region and a budget that excludes the pack; SwapCell interface v0.3; status from PBX-CAL-001
0.3	2026-09-24	Amish Chadha	Removed StepGen as a charging source after StepGen became a walking vehicle (Amish, 2026-09-24)
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

PowerBox requirements

These requirements are checked by calculation in PBX-CAL-001 (TRL 3). They are still proposals, not user-validated needs, and will be revised after co-design sessions. On paper, nine are met, including R2 as restated by Amish on 2026-10-02 (1.87 evenings against 1.8); R6 is at risk, R11 is reported against its value-engineering target (USD 32 over), and R12 cannot be verified until users try a pack swap. The Status column gives the TRL 3 result.

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PBX-CAL-001)
R1	Store useful energy in one SwapCell pack	400 Wh or more usable at the pack terminals, using 10 to 100 % state of charge	Calculation from SwapCell capacity and the host's discharge cut-off	Met, 419 Wh
R2	Run a household through outage evenings	1.8 evenings of the reference load profile (Table 1, 203 Wh at the loads) on one full pack, at the pack's 10 % cut-off (restated from 2 evenings by Amish on 2026-10-02, PBX-DEC-001 item 5)	Energy budget calculation including conversion losses and standby	Met on paper: 1.87 evenings (1.81 with minimum-capacity cells)
R3	Accept four charging sources	(a) DC input 12 to 60 V, up to 200 W, with maximum power point tracking for a solar panel or another DC source; (b) grid through a certified external charger; (c) SunSpoke by swapping in a charged SwapCell pack; (d) SunSpoke's 100 W panel through the DC input	Design review against the input specification table	Met (design review)
R4	Recharge quickly from the grid	10 to 100 % in 3 h or less	Charger rating and SwapCell charge curve	Met, 2.1 h
R5	Recharge from solar in a day	10 to 100 % in one clear day (4.5 peak sun hours) with a 200 W panel	Solar yield calculation	Met, full in 0.70 day
R6	Use DC and solar input efficiently at low power	DC input conversion efficiency 90 % or more from 40 to 200 W, and stable maximum power point tracking (no stall or hunting) as a panel's output changes with light and temperature	Converter efficiency data; later bench test with a solar panel or solar simulator	At risk, 90.5 % worst case in the loss model
R7	Provide the outputs households use	2 x USB-C PD (one 100 W, one 60 W), 2 x USB-A (12 W each), 2 x 12 V DC (10 A shared), 1 x AC outlet, 230 V 50 Hz, 300 W continuous pure sine (THD 3 % or less) behind a 30 mA RCD; total output limited to 400 W	Design review against the output specification table	Met (design review)
R8	Waste little energy when idle	Off: 5 % or less of pack energy per month, including cell self-discharge. Ready (display on, outputs off): 1.0 W or less. Inverter switches itself off after 10 min below 5 W	Standby budget calculation; later measurement	Met, 3.2 % per month off, 0.6 W ready
R9	Never back-feed household wiring	No AC input connector that can mate with an AC output; AC output only through a 30 mA RCD outlet with the inverter neutral bonded to the case; grid charging only through a certified external charger with a DC plug; DC bus 60 V or less	Design review and safety checklist	Met (design review)
R10	Be carried by one adult	10 kg (22 lb) or less with pack; 500 x 300 x 280 mm or smaller including handle; one top handle	Massing model, then weighing	Met, 9.4 kg, 480 x 279 x 275 mm (constructable design)
R11		PowerBox parts against a value-engineering target of USD 450 (a hypothetical control target,	Priced BOM	

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PBX-CAL-001)
	Keep to the value-engineering target	not a limit; Amish, 2026-10-01), including the grid charger and excluding the SwapCell pack, which is priced once in the SwapCell BOM (decided by Amish, 2026-09-25)		USD 482, USD 32 over the target
R12	Swap packs and show status simply	Pack swapped by hand in 30 s or less without tools; display shows state of charge, input and output power and estimated time remaining, read from the SwapCell interface v0.3 CAN messages; a sleeping pack wakes through the INTERLOCK loop (v0.3 item W)	Design review; later timed trial with users	Not verifiable at TRL 3

■ Assumptions

Table 1. Reference evening load profile (6 pm to 11 pm outage). Proposed for review; the real profile must come from co-design sessions.

Load	Power	Hours	Energy at the load	Output used
Three LED bulbs	3 x 5 W	5	75 Wh	12 V DC
Wi-Fi router and fibre terminal	12 W	5	60 Wh	12 V DC
Radio	5 W	4	20 Wh	12 V DC
Four phone charges	about 12 Wh each		48 Wh	USB
Total			203 Wh	

- The SwapCell pack is the reference 13S2P pack of SwapCell interface v0.3: 46.8 V nominal, 39.0 to 54.6 V, 468 Wh nameplate and 466 Wh at 0.2C (SWC-CAL-001). PowerBox is a station host (v0.3 item C) and its receptacle carries the 10 kΩ INTERLOCK coding resistor (item W).
- Solar input spends much of the day well below the panel rating (morning, evening and cloud), so the DC input often runs at 40 to 100 W.
- Refrigerators, kettles and other high-power or high-surge loads are out of scope (see PBX-PRB-001).
- Budget: the USD 450 value-engineering target in `project.yaml` covers the PowerBox itself (decided by Amish, 2026-09-25, PBX-DDR-001). A household that needs its own pack also buys a SwapCell pack, priced once in the SwapCell BOM at about \$414 in prototype parts.
- R2: decided by Amish on 2026-10-02 (PBX-DEC-001, item 5): 1.8 evenings, which covers minimum-capacity cells (1.81); the 10 % cut-off is kept, since discharging to 3.5 % would shorten the shared SwapCell pack's life. The 203 Wh reference evening is not yet validated and is revisited with the first partner's users.